CERN Beamline for Schools Proposal - 2026

Determining Internal Composition of Asteroids Using Muon Tomography Techniques Scaled to Laboratory Conditions

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1 Motivation

Understanding asteroid internal structure is critical for planetary defense and resource utilization, yet current gravity/radar methods lack precision. Sample return missions (Hayabusa2, OSIRIS-REx) cost > 800M USD and return only surface material.

Muon tomography offers a pathway to remote interior mapping, but requires laboratory validation. A NASA NIAC study concluded that "there are presently no established methods to directly characterize the interior structure of an asteroid" and that muography "could provide a direct and cost-effective means of probing the interior density structure" [2]. Our experiment provides the first systematic comparison of asteroid analogue materials using controlled measurements of muon scattering through the same, creating the physics basis for this NASA concept.

Recent work by Ninomiya et al. [1] demonstrated muonic X-ray spectroscopy on Ryugu samples. While they probed *surface composition* via muon capture, our experiment extends this to *internal structure* via muon scattering tomography, a complementary technique essential for full asteroid characterization.

2 Why DESY II Specifically?

DESY II is uniquely suited for this experiment due to five requirements:

- Momentum-selectable electron beam (1–6 GeV/c): Enables verification of $\theta_0 \propto 1/p$ dependence predicted by simulations.
- Single-particle beam with minimal pile-up: DESY's 10^3–10^4 particles/spill rate ensures < 1% pile-up, essential for clean track reconstruction.
- Available Timepix3 detectors with EUDET telescopes: 50 μm resolution enables angular resolution < 0.5 mrad—sufficient to resolve our predicted 0.9–5.8 mrad range.
- Well-characterized beam profile: The 2 cm beam spot (FWHM) matches our 2.5 cm samples, minimizing edge effects.
- Standardized sample holder: < 100 g capacity aligns with our pre-characterized samples (all < 85 g).
- Proven BL4S infrastructure: DESY's experience hosting student experiments ensures productivity.

Note on beam composition: DESY II provides electrons/positrons rather than dedicated muons [9]. However, for our thin samples (1.5 cm thickness, < 0.3 radiation lengths for low-Z materials), multiple Coulomb scattering of 4 GeV electrons follows the same Highland formalism as muons, with scattering angles scaling as $\theta_0 \propto 1/p$. The absence of a dedicated muon beam is mitigated by:

- Our samples' thickness ($\ll$ radiation length) minimizes bremsstrahlung showers
- Electron scattering still enables material discrimination via Z_{eff}
- Previous studies have successfully used electron beams for scattering-based material identification [5, 4]

Thus, DESY's high-energy electron beam effectively mimics cosmic-ray muon scattering for thin samples

3 Theory

The interaction of muons with matter is governed by two fundamental processes. First, energy loss through ionization follows the Bethe-Bloch formula.

$$-\left\langle \frac{dE}{dx} \right\rangle = 2\pi N_a r_e^2 m_e c^2 \rho \frac{Z}{A} \frac{z^2}{\beta^2} \left[\ln\left(\frac{2m_e c^2 \beta^2 \gamma^2 T_{\max}}{I^2} \right) - 2\beta^2 - \delta - \frac{2C}{Z} \right] \tag{1}$$

The multiple Coulomb scattering produces angular deflections described by the Highland formula.

$$\theta_0 = \frac{13.6 \text{ MeV}}{\beta c p} \sqrt{\frac{x}{X_0}} \left[1 + 0.038 \ln\left(\frac{x}{X_0} \right) \right] \tag{2}$$

The radiation length X_0 scales as Z^{-2}, i.e., high-Z materials like Lead produce significantly wider scattering distributions carbonaceous analogues.These equations are implemented in our G4Beamline simulations for accurate GeV-scale muon transport.

4 Hypothesis

We hypothesize that multiple Coulomb scattering of 4 GeV muons can discriminate between asteroid analog materials (carbonaceous, basaltic, metallic) with high statistical significance. Specifically:

Primary Hypothesis

The RMS scattering angle θ_0 will increase monotonically with effective atomic number Z_{eff}, with predicted values (from the Highland formula and our G4Beamline simulations):

- **Carbonaceous** (C-type analog, $Z_{\text{eff}} \sim 5.5$): $\theta_0 = 0.87 \pm 0.02$ mrad
- **Basalt** (S-type analog, $Z_{\text{eff}} \sim 12$): $\theta_0 = 1.95 \pm 0.05$ mrad (composite: SiO_2, Al_2O_3, Fe)
- **Lead** (M-type analog, $Z_{\text{eff}} = 82$): $\theta_0 = 5.77 \pm 0.12$ mrad

These values assume 4 GeV muons and our standard sample geometry (2.5 cm diameter $\times$ 1.5 cm thickness). The ratio $\theta_0^{\text{Pb}}/\theta_0^{\text{C}} \approx 6.6$ provides a clear discrimination signal.

Secondary Hypothesis

The measured scattering angles will match Highland formula predictions within 20%:

$$\left| \frac{\theta_0^{\text{meas}} - \theta_0^{\text{pred}}}{\theta_0^{\text{pred}}} \right| < 0.20 \quad \text{for all materials} \tag{3}$$

Agreement within this bound would validate muon scattering as a reliable technique for material identification, with direct implications for asteroid interior characterization.

Null Hypothesis

Scattering angle distributions are independent of material composition and cannot be distinguished beyond statistical fluctuations. With our planned statistics (10^5 muons per sample), we can reject the null hypothesis if the observed θ_0 values differ by more than 3σ between any two materials.

Statistical Power

With 10^5 muons per sample, the expected significance levels are:

- Lead vs. Carbonaceous: $> 50\sigma$
- Lead vs. Basalt: $> 40\sigma$
- Basalt vs. Carbonaceous: $> 5\sigma$

All material pairs are thus distinguishable at high confidence, confirming the feasibility of laboratory-scale muon tomography for asteroid analog materials.

5 Experimental Setup

To represent the dominant asteroid taxonomic classes, we will study 3 solid analogues with identical geometry and areal density: a basaltic sample (S-type analogue), a Lead sample (M-type analogue), a carbonaceous analogue (C-type). These span an order-of-magnitude variation in density and atomic composition, enabling a systematic study of muon energy loss, multiple scattering, and penetration depth as a function of material properties. Keeping geometry fixed isolates material composition effects from simple mass effects. These three materials were selected to balance astrophysical representativeness with experimental simplicity and limited beam time. The samples are sized to contain the 2 cm FWHM beam spot with alignment margin. and each sample will be characterized for bulk density (ρ), effective Z (Z_{eff}), and elemental composition using XRF analysis at a local university.These parameters define our G4Beamline simulations as input to the G4Material class, enabling direct comparison between predicted and measured scattering distributions.

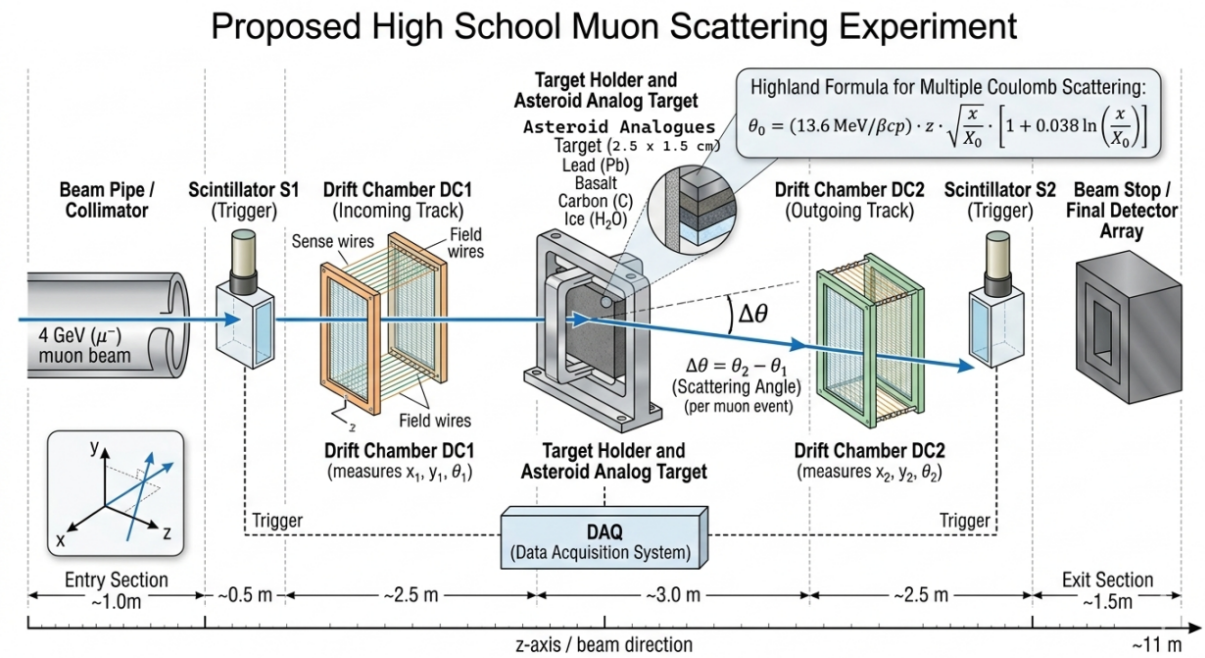


Figure 1: Experimental Setup Diagram

Table 1: Physical and scattering-relevant properties of asteroid analog target materials

Sample	**Asteroid Type**	**Density (g/cm^3)**	Z_{eff}	X_0 **(cm)**	**Primary Physics Effect**
Basalt	S-type (stony)	∼3.0	∼12	∼12–14	Reference case; moderate energy loss and multiple scattering
Lead	M-type (metal-rich)	∼11.3	∼82	∼0.56	Strong stopping power; enhanced multiple scattering
Carbonaceous analogue	C-type (carbon-rich)	∼2.2	∼5.5	∼19–22	Extended muon penetration; low-Z dominated interactions

The quoted Z_{eff} values represent approximate values for composite materials. Radiation lengths X_0 are material-dependent and given here as representative values relevant for high-energy muon transport. All targets: 2.5 cm diameter × 1.5 cm thickness; Simulated Muon beam: 4 GeV, 2 mm FWHM

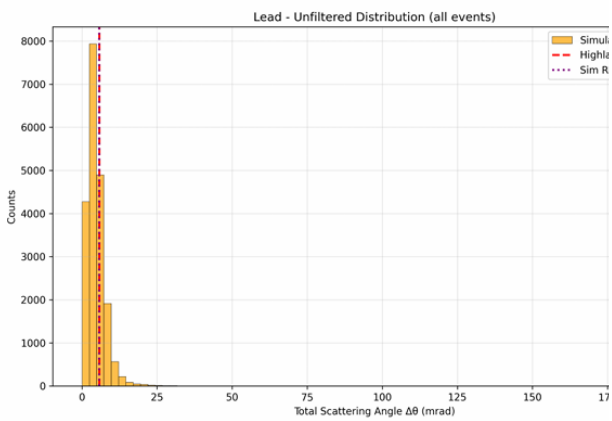
6 PRE - DESY II Work

Before conducting the experiment at DESY II, we modeled all four asteroid analogs on G4Beamline: lead (M-type), basalt (S-type), carbonaceous (C-type), and water ice (cometary/Icy moon analog). Ice was included in simulations to extend the scope of our validation: if our G4Beamline predictions for lead, basalt, and carbonaceous match experimental results at DESY, the ice simulations can be considered validated by proxy, supporting future applications to Icy bodies without requiring additional beamtime.

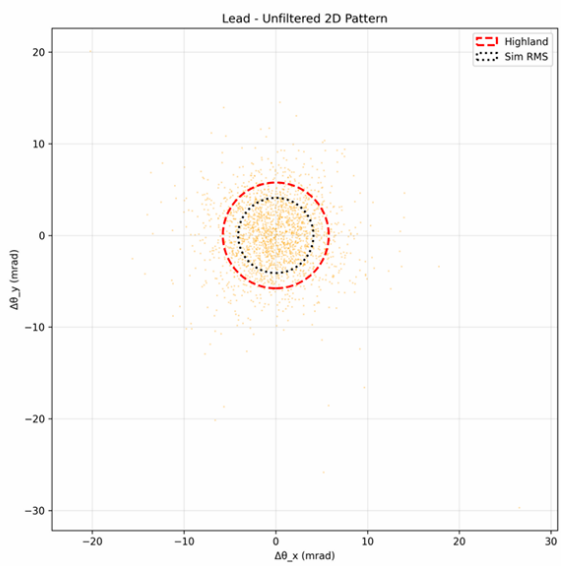
7 Simulations

We performed G4Beamline simulations of 4 GeV muons incident on our three asteroid analog samples (lead, basalt, carbonaceous) plus water ice for comparison. Each simulation tracked 10^5 muons through the exact geometry (2.5 cm diameter $\times$ 1.5 cm thickness) and material compositions listed in Table 1.

Figures 2 - 4 show representative muon tracks through each sample. Figure 5 displays the simulated scattering angle distributions, confirming clear separation between materials: lead (RMS $\theta_0 = 5.8$ mrad) scatters $\sim 6\times$ more than carbonaceous (0.87 mrad), in agreement with Highland formula predictions. These distributions establish our expected signal sizes and inform our statistical projections in Section 8.2.

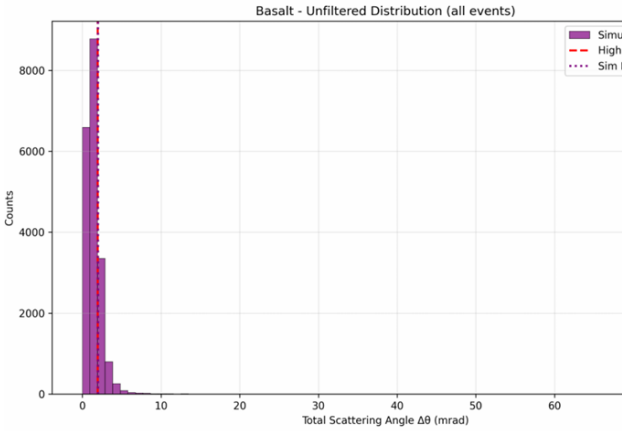


(a) Lead - Distribution

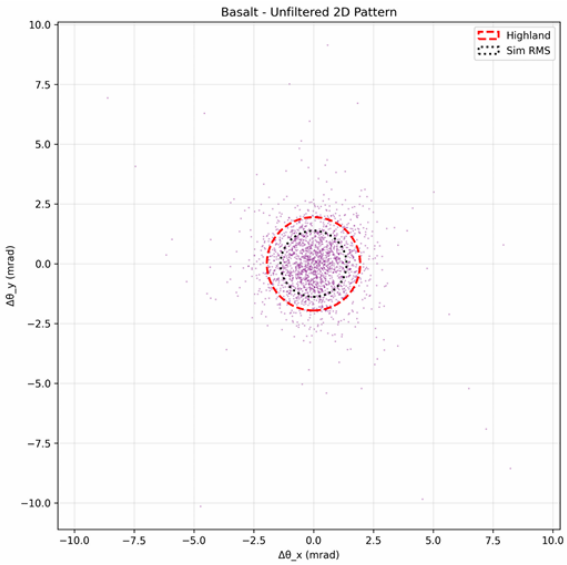


(b) Lead - 2D pattern

Figure 2: Lead

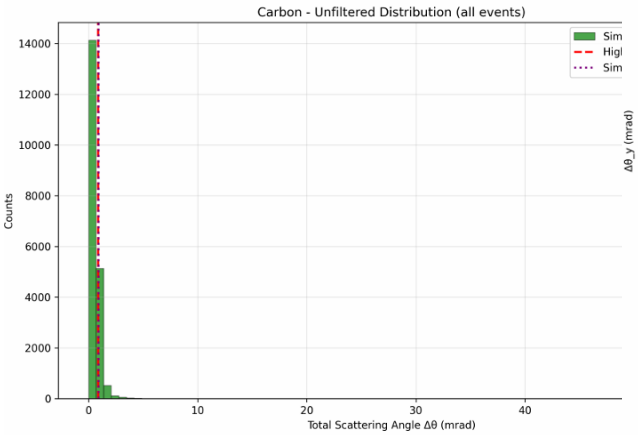


(a) Basalt - Distribution

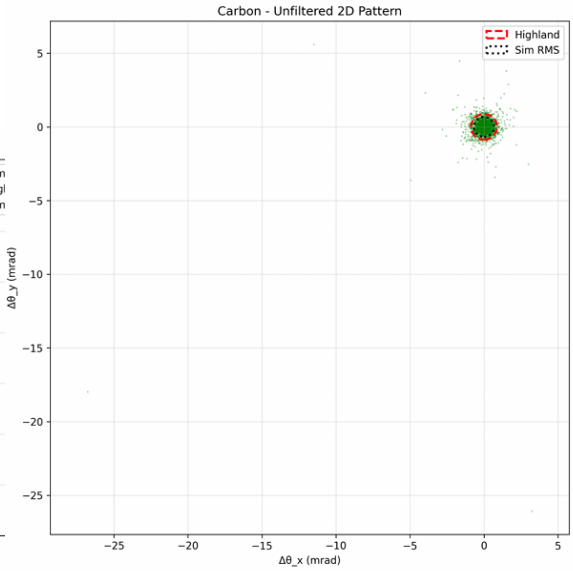


(b) Basalt - 2D pattern

Figure 3: Basalt

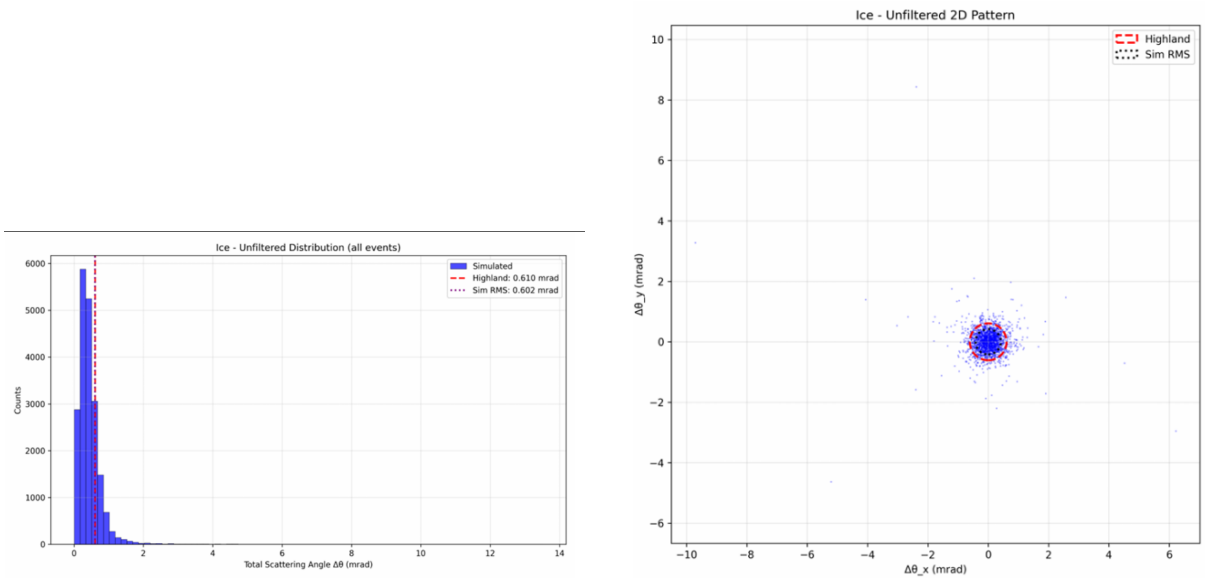


(a) Carbon - Distribution



(b) Carbon - 2D pattern

Figure 4: Carbonaceous (C-type) analog simulation



(a) Ice Core - Distribution

(b) Ice Core - 2D Pattern

Figure 5: Ice

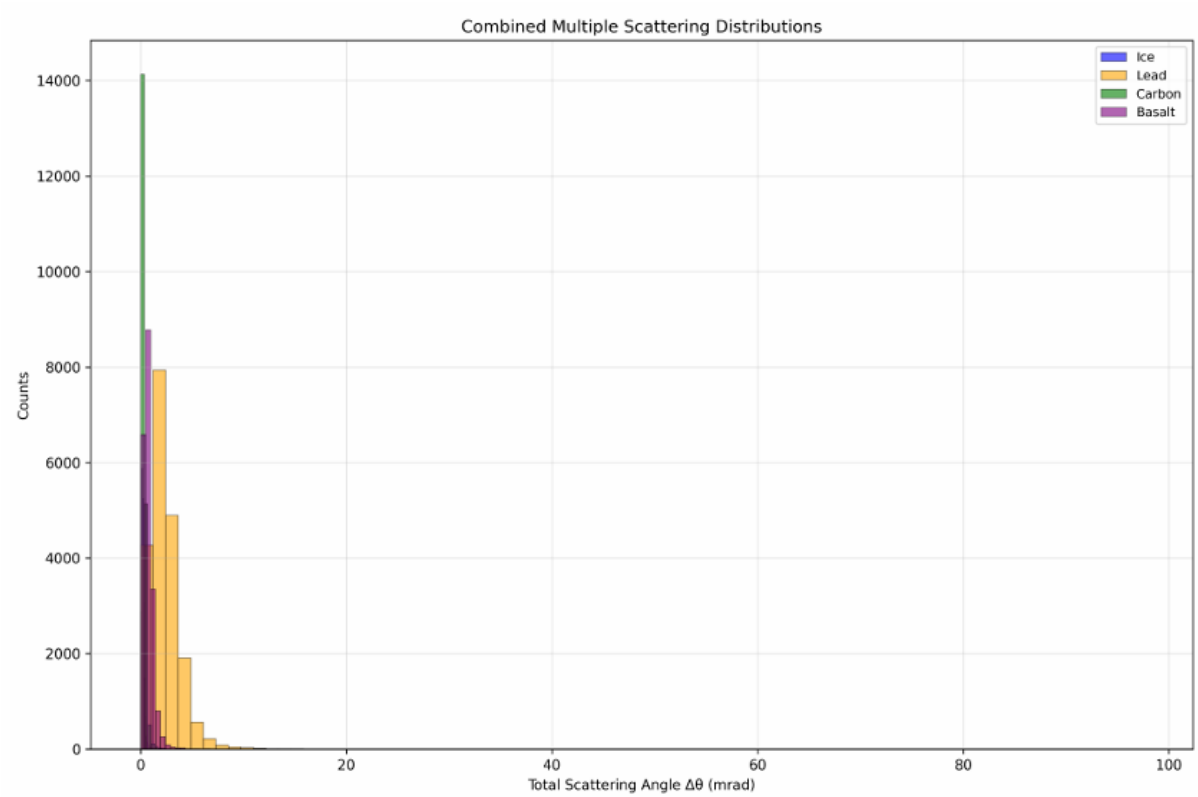


Figure 6: Combined multiple scattering distributions from G4Beamline simulations (lead, basalt, carbonaceous, ice)

Material	Highland Value (mrad)	Simulated Value (mrad)	Simulated % Agreement	Total Used Events	Mean Energy Loss
Carbon	0.87	0.927	106.56%	20000	0.64 MeV
Basalt	1.95	1.9929	102.2%	20000	1.03 MeV
Ice	0.61	0.6017	98.65%	20000	0.37 MeV
Lead	5.77	5.7245	99.21%	20000	2.68 MeV

Table 2: Comparison between Highland prediction and simulation results

8 Beamtime Request and Schedule

We request **32 hours** of beamtime distributed over 6 days within the two-week experimental window. This conservative estimate ensures sufficient statistics while accounting for setup, calibration, and contingency.

8.1 Beamtime Allocation

Table 3: Detailed 32-hour beamtime schedule

Day	**Activity**	**Hours**	**Cumulative**
Day 1	**Setup and Calibration**		
	• Transport equipment to experimental area	1.0	1.0
	• Install sample holder and verify positioning system	1.0	2.0
	• Detector alignment (Timepix3 telescopes)	2.0	4.0
	• High voltage optimization and trigger configuration	1.0	5.0
	• Beam tuning (4 GeV, verify profile)	1.0	6.0
	• DAQ test runs with open beam	1.0	7.0
Day 2	**Lead Measurements**		
	• Install lead sample, verify alignment	0.5	7.5
	• Data collection: 10^5 events ($\sim$4 hours at 8 Hz)	4.0	11.5
	• Preliminary online monitoring, verify scattering	1.0	12.5
	• Background run (open beam, 30 min)	0.5	13.0
Day 3	**Basalt Measurements**		
	• Swap to basalt sample, verify alignment	0.5	13.5
	• Data collection: 10^5 events	4.0	17.5
	• Preliminary analysis, compare to G4Beamline	1.0	18.5
	• Position variation study (off-center, 30 min)	0.5	19.0
Day 4	**Carbonaceous Measurements**		
	• Swap to carbon sample, verify alignment	0.5	19.5
	• Data collection: 10^5 events	4.0	23.5
	• Preliminary analysis, compare to predictions	1.0	24.5
	• Background run (open beam, 30 min)	0.5	25.0
Day 5	**Systematic Studies**		
	• Energy dependence (2 GeV and 6 GeV on lead sample)	2.0	27.0
	• Repeat critical measurements (basalt, 30 min)	0.5	27.5
	• Detector alignment check	0.5	28.0
	• Contingency time for re-runs	2.0	30.0
Day 6	**Analysis and Dismantling**		
	• Final cross-checks and data integrity verification	1.0	31.0
	• Equipment removal and cleanup	1.0	32.0
		Total Beamtime	**32 hours**

8.2 Statistical Justification

Our statistics estimate assumes:

- Average event rate: 8 Hz (conservative, based on DESY specifications for 4 GeV)
- Data collection efficiency: 80% (dead time, triggering)
- Hours per sample: 4 hours $\times 3600$ s/hr $\times 8$ Hz $\times 0.8 \approx 9.2 \times 10^4$ muons per sample
- Total per sample: $\sim 10^5$ events, sufficient for $< 1\%$ statistical uncertainty on θ_0

With 10^5 events per material, the statistical separation between samples is approximately 270 σ

Even the smallest difference (basalt vs. carbonaceous, $\Delta\theta \approx 1.08$ mrad) yields $> 5\sigma$ separation.

8.3 Contingency Planning

The 2-hour contingency block on Day 5 accommodates:

- Unexpected beam interruptions
- Additional alignment checks if needed
- Repeat of any sample where data quality flags issues
- Extra energy points if ahead of schedule

This schedule is deliberately conservative, prioritizing data quality over quantity. All critical measurements (three samples, open beam backgrounds, energy dependence) are achievable within 32 hours, with margin for unexpected delays.

9 Data Acquisition and Analysis

For each event, we reconstruct incoming/outgoing tracks from telescope hits to determine scattering angle and lateral displacement. We build statistical distributions for each material and compare to:

- Highland formula predictions using our independently measured x/X_0
- G4Beamline simulations with identical geometry

The RMS scattering angle θ_0 for each material will be extracted via Gaussian fits to the core of the distribution. Discrepancies beyond 20% will indicate detector effects or additional physics not captured by the simple model.

9.1 Safety and Technical Considerations

This experiment is designed to be conducted strictly within the available equipment at DESY, falling well within safety norms and logistical capabilities of the facility.
Sample Safety:

- All samples will be solid, non-radioactive, non-toxic
- Maximum weight $<$ 85g each
- Mechanically simple and modular
- Data volumes are moderate and compatible with typical beam-telescope operations
- BL4S-provided Timepix detectors and standard trigger scintillators are used
- No custom high-voltage or vacuum systems required
- Samples can be easily swapped (5-minute procedure)

Beam Compatibility:

- Samples fit within DESY's standard sample holder
- No interference with beamline diagnostics
- Radiation levels within sample activation limits (checked via simulation)

9.2 Expected Results

Minimum Viable Result:

- Statistically significant difference in scattering between high-Z (Lead) and low-Z (carbonaceous) materials and validation of G4Beamline predictions within 20%.

Optimal Result:

- Complete discrimination between all three materials with $> 5\sigma$ confidence
- Measurement of attenuation lengths matching theoretical predictions
- Scalability demonstration: Show how lab data can inform full-scale asteroid missions

10 Future work and Astrophysical Scaling

- Future work could reconstruct density distributions using voxel-based muon scattering tomography algorithms.
- Because scattering variance scales with material thickness and density, these laboratory measurements can be extrapolated to kilometer-scale asteroid interiors using established muon transport models.

11 What We Hope to Take Away

BL4S is more than a physics experiment, it is a gateway to international collaboration and scientific citizenship.

Scientific Impact: Our experiment will provide the first controlled measurements of muon scattering through well-characterized asteroid analogs at GeV energies, creating a laboratory foundation for NASA's concept of asteroid muography [2]. These data will be made publicly available for future researchers.

Personal Growth: Our team spans three countries (Egypt, India, USA) and diverse backgrounds, different members worked on different tasks - physics calculations/theory, G4Beamline simulations, contacting professors, literature/writing. BL4S offers the rare opportunity to work as a unified team across these differences, mirroring the international collaboration at the heart of facilities like DESY. Working with mentors from IIT Kharagpur and Johannes Kepler University has prepared us to think like researchers.

12 Outreach and Community Impact

BL4S is more than a physics experiment — it is a gateway to international collaboration and scientific citizenship. Our outreach plan extends beyond the beamtime:

Pre-DESY Engagement: We have already begun sharing our journey through school presentations and social media, building excitement about particle physics in our local communities in Egypt, India, and the USA.

Post-Experiment Commitments:

- **Workshops:** Develop a muon physics workshop for middle-school students using our cosmic ray detector prototype and cloud chamber as demonstration tools.
- **Open Science:** Publish our results, analysis code, and a video series documenting the experiment in formats accessible to future BL4S teams and student researchers.
- **International Network:** Create a lasting collaboration connecting students across Egypt, India, and the USA to encourage future cross-border science projects.
- **Academic Dissemination:** Present our findings at regional science fairs and submit to junior science journals. Several team members plan to apply to physics programs inspired by this experience.

Previous winners describe BL4S as transformative, teaching that "science is not only about physics — it is engineering, IT, and communication" [12]. We seek that same transformation and commit to carrying its spirit back to our communities

13 Acknowledgments

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